

Job Referral System

24BCE10110

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Introduction

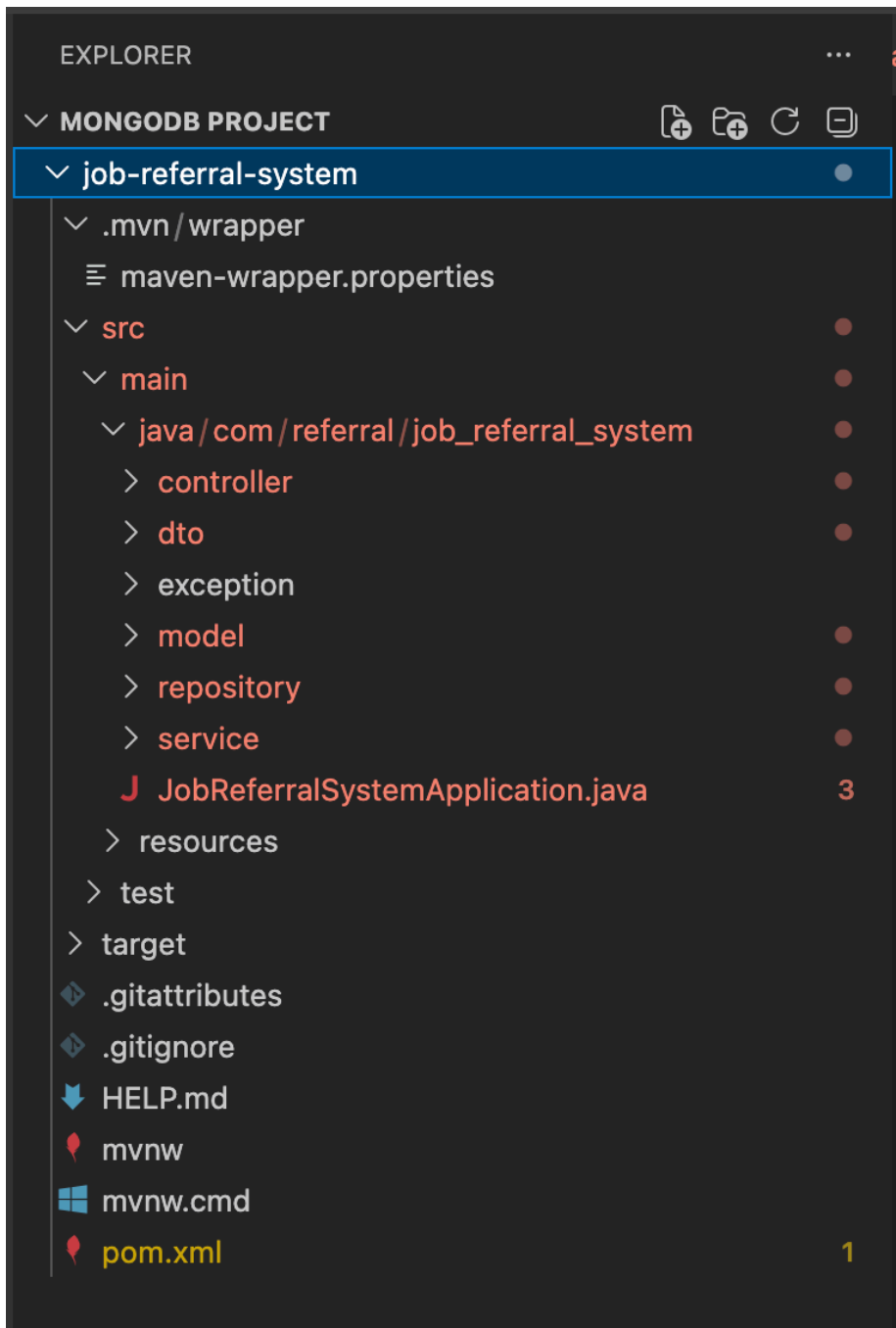
The Job Referral System is a backend application developed using Spring Boot and MongoDB. The project helps manage companies, job openings, and candidate referrals through REST APIs. All APIs were tested using Swagger UI.

Tools Used

- Java 17
- Spring Boot 3.x
- MongoDB 8.x
- Maven 3.x
- Swagger UI (OpenAPI 3.1)
- VS Code
- Git & GitHub

Execution and Output Screenshots

Project Structure



This screenshot shows the project structure including controller, service, repository, DTO, model, and exception packages.

Swagger UI

Swagger UI interface showing the OpenAPI definition for a referral-controller, job-controller, and company-controller. The interface includes a search bar, a list of endpoints, and a schema section.

OpenAPI definition v0 OAS 3.1

Servers
http://localhost:8080 - Generated server url

referral-controller

- PUT /referrals/{id}
- DELETE /referrals/{id}
- GET /referrals
- POST /referrals
- GET /referrals/job/{jobId}

job-controller

- PUT /jobs/{id}
- DELETE /jobs/{id}
- GET /jobs
- POST /jobs
- GET /jobs/company/{companyId}

company-controller

- GET /companies/{id}
- PUT /companies/{id}
- DELETE /companies/{id}
- GET /companies
- POST /companies
- GET /companies/test
- GET /companies/all

Schemas

- Referral > Expand all object
- JobRequestDto > Expand all object
- Job ^ Collapse all object
 - id string
 - title string
 - description string
 - location string
 - companyId string
- Company > Expand all object

Swagger UI was used to test all REST APIs for Company, Job, and Referral modules.

MongoDB Collections

```
abhiyeet@Abhiyeets-MacBook-Air MongoDB project % mongosh
test 192.00 KiB
<database_name>> use test
switched to db test
test> show collections
companies
jobs
referrals
test>
```

MongoDB collections created for this project are companies, jobs, and referrals.

Create Company

The screenshot shows a REST client interface with the following sections:

- POST /companies**: The endpoint and method.
- Parameters**: No parameters are present.
- Request body**: A JSON object:

```
{  "companyName": "Google India",  "location": "Pune",  "industry": "Software"}

```
- Execute**: A blue button to send the request.
- Responses**:
 - Curl**: A terminal window showing the curl command used to send the request.
 - Request URL**: `http://localhost:8888/companies`
 - Server response**:
 - Code**: 200
 - Response body**: A JSON object:

```
{  "id": "506c70a07886c510c040",  "companyName": "Google India",  "location": "Pune",  "industry": "Software",  "jobIds": null}

```
 - Response headers**:

```
connection: keep-alive
content-type: application/json
date: Mon, 15 Jun 2026 16:23:39 GMT
keep-alive: timeout=5
transfer-encoding: chunked

```
- Responses**: A table with the following columns: Code, Description, Links.

Code	Description	Links
200	OK	No links
- Media type**: A dropdown menu showing `*/*`.
- Content Accept header**: A text input field.
- Example Value**: A JSON schema:

```
{  "id": "string",  "companyName": "string",  "location": "string",  "industry": "string",  "jobIds": [    "string"  ]}

```

A new company record was created successfully and stored in MongoDB.

Get Companies

GET /companies/all

Parameters

No parameters

Execute Clear

Responses

Curl

curl -X 'GET' \n'http://localhost:8080/companies/all' \n-H 'accept: */*'

Request URL

http://localhost:8080/companies/all

Server response

Code

Details

200

Response body

{\n \"id\": \"6c38279b078405d51ed5&c3\", \n \"companyName\": \"Google India\", \n \"location\": \"Pune\", \n \"industry\": \"Software\", \n \"jobIds\": null \n}

Response headers

connection: keep-alive \ncontent-type: application/json \ndate: Mon, 15 Jun 2026 16:24:42 GMT \nkeep-alive: timeout=60 \ntransfer-encoding: chunked

Responses

Code

Description

Links

200

OK

No links

Media type

/

Controls Accept header.

Example Value Schema

{\n \"id\": \"string\", \n \"companyName\": \"string\", \n \"location\": \"string\", \n \"industry\": \"string\", \n \"jobIds\": [\n \"string\" \n] \n}

All company records were retrieved successfully.

Update Company

PUT

/companies/{id}

Cancel

Reset

Name	Description
id required	
string (path)	6a30270b078405d511ed58c3

Request body required

application/json

```
{  "companyName": "Google India",  "location": "Mumbai",  "industry": "Software"}  
```

ExecuteClear

Responses

Curl

```
curl -X 'PUT' \  "http://localhost:8080/companies/6a30270b078405d511ed58c3" \  -H 'accept: */*' \  -H 'Content-Type: application/json' \  -d '{  "companyName": "Google India",  "location": "Mumbai",  "industry": "Software"}  '
```

Request URL

http://localhost:8080/companies/6a30270b078405d511ed58c3

Server response

Code	Details
200	Response body <pre>{ "id": "6a30270b078405d511ed58c3", "companyName": "Google India", "location": "Mumbai", "industry": "Software", "jobsIds": null }</pre> Download Response headers <pre>connection: keep-alive content-type: application/json date: Sun, 15 Jul 2018 18:41:34 GMT keep-alive: timeout=60 transfer-encoding: chunked </pre>

Responses

Code	Description	Links
200	OK	No links

Media type

/

Control Accept header

Example Value | Schema

```
{  "id": "string",  "companyName": "string",  "location": "string",  "industry": "string",  "jobsIds": [    "string"  ]  }
```

Company information was updated successfully.

Delete Company

company-controller

GET /companies/{id}

PUT /companies/{id}

DELETE /companies/{id}

Cancel

Parameters

Name	Description
id required	
string	
(path)	

Execute

Clear

Responses

Curl

curl -X 'DELETE' \n'http://localhost:8080/companies/6a30270b078405d511ed58c3' \n-H 'accept: */*'

Request URL

http://localhost:8080/companies/6a30270b078405d511ed58c3

Server response

Code	Details
200	Response body Company deleted successfully Response headers connection: keep-alive content-length: 28 content-type: text/plain; charset=UTF-8 date: Mon, 15 Jun 2026 17:26:09 GMT keep-alive: timeout=60

Responses

Code	Description	Links
200	OK	No links

Media type

/

Controls Accept header.

Example Value | Schema

string

Selected company record was deleted successfully.

Create Job

POST

/jobs

Parameters

No parameters

Request body required

application/json

```
{  "title": "Software Engineer",  "description": "Java Spring Boot Developer",  "companyId": "6a38278b87485b6d511ed58c3"}  
```

Execute

Clear

Responses

Curl

```
curl -X 'POST' \  "http://localhost:8080/jobs" \  -H 'accept: */*' \  -H 'Content-Type: application/json' \  -d '{  "title": "Software Engineer",  "description": "Java Spring Boot Developer",  "companyId": "6a38278b87485b6d511ed58c3"}'  
```

Request URL

http://localhost:8080/jobs

Server response

Code	Details
200	Response body <pre>{ "id": "6a382c69874485d511ed58c5", "title": "Software Engineer", "description": "Java Spring Boot Developer", "location": null, "companyId": "6a38278b87485b6d511ed58c3"} </pre> Download Response headers <pre>connection: keep-alive content-type: application/json date: Mon, 15 Jun 2026 16:46:33 GMT keep-alive: timeout=60 transfer-encoding: chunked </pre>

Responses

Code	Description	Links
200	OK	No links

Media type

/

Controls Accept header.

Example Value

Schema

```
{  "id": "string",  "title": "string",  "description": "string",  "location": "string",  "companyId": "string"}  
```

A job opening was created and linked to a company.

Get Jobs

job-controller

PUT	/jobs/{id}	>
DELETE	/jobs/{id}	>
GET	/jobs	>

Parameters Cancel

No parameters

Execute

Responses

Curl

```
curl -X GET \
'http://localhost:8080/jobs' \
-H 'accept: */*''
```

Request URL

```
http://localhost:8080/jobs
```

Server response

Code	Details
200	Response body <pre>{ { "id": "6a302c28070405d511ed58c4", "title": "Software Engineer", "description": "Java Spring Boot Developer", "location": null, "companyId": "PASTE_COMPANY_ID" }, { "id": "6a302c09070405d511ed58c5", "title": "Software Engineer", "description": "Java Spring Boot Developer", "location": null, "companyId": "6a30270b070405d511ed58c3" } }</pre> Response headers <pre>connection: keep-alive content-type: application/json date: Mon, 15 Jun 2020 10:49:20 GMT keep-alive: timeout=60 transfer-encoding: chunked</pre>

Responses

Code	Description	Links
200	OK	No links

Media type

/

Controls Accept header

Example Value | Schema

```
{
  {
    "id": "string",
    "title": "string",
    "description": "string",
    "location": "string",
    "companyId": "string"
  }
}
```

All available job records were displayed.

Update Job

job-controller

PUT /jobs/{id}

CancelReset

Parameters

Name	Description
id required	
string	
(path)	

Request body required

application/json

```
{  "title": "Senior Software Engineer",  "description": "Spring Boot Developer",  "companyId": "6a302c28078405d511ed58c4"}  
```

Execute

Clear

Responses

Curl

```
curl -X 'PUT' \  "http://localhost:8080/jobs/6a302c28078405d511ed58c4" \  -H 'Accept: */*' \  -H 'Content-Type: application/json' \  -d '{  "title": "Senior Software Engineer",  "description": "Spring Boot Developer",  "companyId": "6a302c28078405d511ed58c4"}'  
```

Request URL

http://localhost:8080/jobs/6a302c28078405d511ed58c4

Server response

Code	Details
200	Response body <pre>{ "id": "6a302c28078405d511ed58c4", "title": "Senior Software Engineer", "description": "Spring Boot Developer", "location": null, "companyId": "6a302c28078405d511ed58c4"} </pre> Download Response headers <pre>connection: keep-alive content-type: application/json date: Mon, 15 Jun 2026 16:53:01 GMT keep-alive: timeout=6 transfer-encoding: chunked </pre>

Responses

Code	Description	Links
200	OK	No links

Media type

/

Controls Accept header.

Example Value

Schema

```
{  "id": "string",  "title": "string",  "description": "string",  "location": "string",  "companyId": "string"}  
```

Job details were updated successfully.

Delete Job

job-controller

PUT /jobs/{id}

DELETE /jobs/{id}

Parameters

Name	Description
id required	
string (path)	6a302c69078405d511ed58c5

ExecuteClear

Responses

Curl

curl -X 'DELETE' \n "http://localhost:8080/jobs/6a302c69078405d511ed58c5" \n -H 'accept: */*'

Request URL

http://localhost:8080/jobs/6a302c69078405d511ed58c5

Server response

Code	Details
200	Response body Job deleted successfully Response headers connection: keep-alive\n content-length: 24\n content-type: text/plain; charset=UTF-8\n date: Mon, 15 Jun 2026 17:27:18 GMT\n keep-alive: timeout=60

Responses

Code	Description	Links
200	OK	No links

Media type

/

Content Accept header

Example Value | Schema

string

Selected job record was removed successfully.

Create Referral

referral-controller

PUT /referrals/{id}

DELETE /referrals/{id}

GET /referrals

POST /referrals

Parameters

Cancel

Reset

No parameters

Request body required

application/json

```
{
  "candidateName": "Abhijeet Patil",
  "candidateEmail": "abhij@gmail.com",
  "jobId": "PASTE_JOB_ID",
  "referrerName": "Rahul",
  "status": "PENDING"
}
```

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
'http://localhost:8080/referrals' \
-H 'accept: */*' \
-H 'Content-Type: application/json' \
-d '{
  "candidateName": "Abhijeet Patil",
  "candidateEmail": "abhij@gmail.com",
  "jobId": "PASTE_JOB_ID",
  "referrerName": "Rahul",
  "status": "PENDING"
}'
```

Request URL

http://localhost:8080/referrals

Server response

Code	Details
200	Response body <pre>{ "id": "6a382e73878485d51ed58c6", "candidateName": "Abhijeet Patil", "candidateEmail": "abhij@gmail.com", "jobId": "PASTE_JOB_ID", "referrerName": "Rahul", "status": "PENDING" }</pre> Download Response headers <pre>connection: keep-alive content-type: application/json date: Mon, 15 Jun 2026 16:55:15 GMT keep-alive: 120000 transfer-encoding: chunked</pre>

Responses

Code	Description	Links
200	OK	No links

Media type

/

Content Accept header

Example Value | Schema

```
{
  "id": "string",
  "candidateName": "string",
  "candidateEmail": "string",
  "jobId": "string",
  "referrerName": "string",
  "status": "string"
}
```

A candidate referral record was created successfully.

Get Referrals

referral-controller

PUT /referrals/{id}

DELETE /referrals/{id}

GET /referrals

Parameters

No parameters

Execute

Clear

Responses

Curl

curl -X 'GET' \n 'http://localhost:8080/referrals' \n -H 'accept: */*'

Request URL

http://localhost:8080/referrals

Server response

Code

Details

200

Response body

[
 {
 "id": "6a30229bcofde81c48ca82f",
 "candidateName": "Abhijeet Patil",
 "candidateEmail": "abhi@gmail.com",
 "jobId": "6a300420cc0be18be62bede9",
 "refererName": "Rahul",
 "status": "PENDING"
 },
 {
 "id": "6a302e73078495d511e458c6",
 "candidateName": "Abhijeet Patil",
 "candidateEmail": "abhi@gmail.com",
 "jobId": "PASTE_JOB_ID",
 "refererName": "Rahul",
 "status": "PENDING"
 }
]

Response headers

connection: keep-alive
content-type: application/json
date: Mon, 15 Jun 2026 16:59:53 GMT
keep-alive: timeout=60
transfer-encoding: chunked

Responses

Code

Description

Links

200

OK

No links

Media type

/

Content Accept header

Example Value | Schema

[
 {
 "id": "string",
 "candidateName": "string",
 "candidateEmail": "string",
 "jobId": "string",
 "refererName": "string",
 "status": "string"
 }
]

All referral records were displayed.

Update Referral

referral-controller

PUT /referrals/{id}

Cancel

Reset

Parameters

Name	Description
id required	
string (path)	6a302e73078405d511ed58c6

Request body required

application/json

```
{  "candidateName": "Abhijeet Patil",  "candidateEmail": "abhijeet@gmail.com",  "jobId": "6a302e69878405d511ed58c5",  "referrerName": "Rahul",  "status": "APPROVED"}
```

Execute

Clear

Responses

Curl

curl -X 'PUT' \ "http://localhost:8080/referrals/6a302e73078405d511ed58c6" \ -H 'Accept: */*' \ -H 'Content-Type: application/json' \ -d '{ "candidateName": "Abhijeet Patil", "candidateEmail": "abhijeet@gmail.com", "jobId": "6a302e69878405d511ed58c5", "referrerName": "Rahul", "status": "APPROVED" }'

Request URL

http://localhost:8080/referrals/6a302e73078405d511ed58c6

Server response

Code	Details
200	Response body <pre>{ "id": "6a302e73078405d511ed58c6", "candidateName": "Abhijeet Patil", "candidateEmail": "abhijeet@gmail.com", "jobId": "6a302e69878405d511ed58c5", "referrerName": "Rahul", "status": "APPROVED"}</pre> Download Response headers connection: keep-alive content-type: application/json date: Mon, 15 Jun 2026 17:41:51 GMT keep-alive: timeout=60 transfer-encoding: chunked

Responses

Code	Description	Links
200	OK	No links

Media type

/

Content Accept header

Example Value | Schema

```
{  "id": "string",  "candidateName": "string",  "candidateEmail": "string",  "jobId": "string",  "referrerName": "string",  "status": "string"}
```

Referral details and status were updated successfully.

Delete Referral

The image shows a Swagger UI interface for a DELETE endpoint. The endpoint is `DELETE /referrals/{id}`. The parameter `id` is a required string (path) with the value `6a302e73078405d511ed58c6`. The response is a 200 status code with the body `Referral deleted successfully`. The response headers are `connection: keep-alive`, `content-length: 29`, `content-type: text/plain; charset=UTF-8`, `date: Mon, 15 Jun 2020 17:00:08 GMT`, and `keep-alive: timeout=60`. The response is also shown in a table with the code `200`, description `OK`, and media type `*/*`.

Code	Description	Links
200	OK	No links

Referral record was deleted successfully.

Outcome

Through this project, I learned how to build REST APIs using Spring Boot, connect applications with MongoDB, perform CRUD operations, and test APIs using Swagger UI.

Summary

This project demonstrates a simple job referral management system. Companies, jobs, and referrals can be managed efficiently through backend APIs.

Conclusion

The Job Referral System was successfully implemented and tested. All CRUD operations worked correctly, and data was stored and retrieved from MongoDB successfully. This project improved my understanding of backend development using Spring Boot and MongoDB.

~Thank you